

Technical report: allometric carbon allocation with labile storage and validation tests

1. Purpose of this report

This report documents the current allometric carbon allocation scheme implemented in `carbon_allocation_offline_storage_with_checks.f90` and summarizes the validation tests performed before adding growth respiration. The objective was to verify whether the new gradual allocation routine with a labile storage pool is numerically stable, conserves carbon, and behaves consistently under a broad range of physiological and allometric conditions.

The report focuses on the routine currently used in the storage-based offline tests:

```
allocate_gradual_with_storage(...)
```

The legacy routines `allocate(...)` and `abnormal_allocation(...)` are still present in the module, but they are not called by the current storage-based test drivers. Therefore, the tests described here validate the gradual storage allocation pathway, not the legacy rigid allocation pathway.

2. Conceptual rationale of the allocation scheme

The new allocation scheme was designed to represent carbon allocation at a daily timestep without forcing the plant to instantaneously satisfy all allometric constraints. In the previous rigid allocation logic, all available carbon was allocated immediately to leaf, fine root, and sapwood compartments in a way that attempted to close the allometric equations at the end of the timestep. This type of formulation is mathematically convenient, but it is biologically strong for daily allocation because real plants do not instantaneously reorganize their architecture every day.

The gradual storage-based scheme follows a different logic:

1. Daily NPP enters a labile carbon storage pool.
2. Structural growth demand is computed from two components:
 - allometric correction demand, which moves the plant toward leaf-root and pipe-model balance;
 - background balanced-growth demand, which allows continued structural growth even when the plant is already close to allometric equilibrium.
3. Carbon is withdrawn from storage and allocated to structural pools only up to the minimum of:
 - available storage after daily NPP input;
 - total structural demand;
 - a maximum daily allocation limit.

4. The allocated carbon is distributed proportionally among leaf, fine root, and sapwood according to their relative demands.
5. Height is updated from total stem carbon rather than by forcing the pipe-model residual to zero at each daily timestep.

This formulation makes allocation sink-limited rather than purely source-driven. It allows temporary carbon accumulation in storage and gradual convergence toward allometric consistency.

3. Carbon pools and state variables

The current test implementation tracks the following plant carbon compartments:

Compartment	Role in the allocation scheme
<code>leaf_mass</code>	Structural carbon in leaves. Determines leaf area through SLA.
<code>root_mass</code>	Structural carbon in fine roots. Constrained by the target leaf-root ratio.
<code>sapwood_mass</code>	Conductive stem carbon. Used in the pipe-model relationship.
<code>heartwood_mass</code>	Non-living or inactive stem carbon. It is kept constant in the current gradual allocation routine.
<code>carbon_storage</code>	Labile carbon pool that receives daily NPP and pays for structural allocation.
<code>height</code>	Recomputed from total stem carbon using the height-diameter/allometric relationship.

At this stage, storage is a labile carbon pool but does not yet include explicit storage respiration, mobilization costs, or reserve turnover. Growth respiration is also not included yet.

4. Main allometric constraints

4.1 Leaf-root balance

The target leaf-root relationship is:

$$\text{leaf_mass} = \text{leaf_to_root_ratio} \times \text{root_mass}$$

In the base test case, `leaf_to_root_ratio = 1`, so the target is approximately:

```
leaf_mass = root_mass
```

The diagnostic residual is:

```
leaf_root_residual = leaf_mass - leaf_to_root_ratio * root_mass
```

A positive residual indicates that leaf mass is high relative to root mass. A negative residual indicates that root mass is high relative to leaf mass.

4.2 Pipe-model relationship

The pipe model links leaf area and sapwood area:

```
leaf_area = latosa * sapwood_area
```

Leaf area is computed from leaf carbon and SLA:

```
leaf_area = leaf_mass * SLA
```

Sapwood area is derived from sapwood mass, wood density, and height:

```
sapwood_area = sapwood_mass / (wood_density * height)
```

The pipe-model residual is therefore:

```
pipe_residual = leaf_area - latosa * sapwood_area
```

A negative pipe residual indicates that leaf area is too low relative to the existing sapwood area. A positive pipe residual indicates that leaf area is high relative to sapwood area.

4.3 Height update

In the gradual allocation routine, height is not adjusted to force the pipe model to close instantly. Instead, height is recomputed from total stem carbon:

```
total_stem_carbon = sapwood_mass + heartwood_mass
```

This prevents unrealistic daily height jumps and allows pipe-model residuals to decline gradually through structural allocation.

5. Daily allocation algorithm

5.1 Daily NPP input to storage

Annual or rate-based NPP input is converted to daily carbon input using:

```
npp_daily = npp_rate * dt_years
```

With a daily timestep:

```
dt_years = 1 / 365
```

Daily NPP is first added to labile storage:

```
storage_after_npp = carbon_storage_before + npp_daily
```

If daily NPP is negative and exceeds available storage, storage is clamped at zero and the remaining deficit is reported as:

```
unmet_storage_deficit
```

This makes the carbon deficit explicit rather than hiding it inside a storage clamp.

5.2 Allometric correction demand

The routine computes target structural states based on the current allometric imbalance. The plant is not forced to reach these targets immediately. Instead, the difference between current and target values is converted into a daily correction demand using:

```
daily_correction_demand = allometric_deficit / allometric_adjustment_days
```

The parameter `allometric_adjustment_days` therefore controls how rapidly the plant moves toward allometric balance.

5.3 Background balanced-growth demand

If the plant is already close to allometric equilibrium, pure correction-based allocation can become very small. In that case, positive NPP may accumulate indefinitely in storage even though a biologically realistic plant should continue growing in a balanced way.

To avoid this behavior, the routine includes background balanced-growth demand:

```
leaf_background_demand = leaf_target * dt_years / leaf_background_timescale_years  
root_background_demand = root_target * dt_years / root_background_timescale_years  
sapwood_background_demand = sapwood_target * dt_years / sapwood_background_timescale_years
```

In the tested background-growth mode, the timescales were:

Pool	Background growth timescale
Leaf	3 years
Fine root	3 years
Sapwood	15 years

When background growth is disabled, these timescales are set to zero and allocation depends only on explicit allometric correction deficits.

5.4 Daily allocation limiter

The total amount of carbon that can be allocated to structural growth in one timestep is limited by:

```
carbon_to_allocate = min(storage_after_npp, total_demand_daily, max_daily_allocation)
```

where:

```
max_daily_allocation = max_allocation_fraction * living_carbon
```

and:

```
living_carbon = leaf_mass + root_mass + sapwood_mass
```

This prevents unrealistically large structural jumps in a single daily timestep.

5.5 Proportional distribution among structural pools

Once the total daily structural allocation is determined, carbon is distributed proportionally to the relative demand of each pool:

```
frac_leaf    = leaf_demand_daily / total_demand_daily
frac_root    = root_demand_daily / total_demand_daily
frac_sapwood = sapwood_demand_daily / total_demand_daily
```

Then:

```
delta_leaf    = frac_leaf * carbon_to_allocate
delta_root    = frac_root * carbon_to_allocate
delta_sapwood = frac_sapwood * carbon_to_allocate
```

The structural pools are updated by adding these increments. Heartwood is unchanged in the current gradual allocation routine.

6. Carbon-accounting diagnostics added to the routine

Before adding growth respiration, explicit carbon-accounting diagnostics were added to the allocation output. These diagnostics are essential because growth respiration will introduce an additional carbon sink, and the allocation scheme must be demonstrably conservative before adding that process.

6.1 Structural carbon balance

This diagnostic checks whether all carbon assigned to structural growth was actually distributed among leaf, root, and sapwood increments:

```
structural_balance_error = carbon_to_allocate - (delta_leaf + delta_root + delta_sapwood)
```

Expected value:

```
structural_balance_error  0
```

6.2 Storage carbon balance

This diagnostic checks whether final storage follows from initial storage, daily NPP, any explicit unmet deficit, and structural allocation:

```
storage_balance_error = carbon_storage_after -  
    (carbon_storage_before + npp_daily + unmet_storage_deficit - carbon_to_allocate)
```

Expected value:

```
storage_balance_error    0
```

6.3 Whole-plant carbon balance

This diagnostic checks the combined change in structural carbon plus storage:

```
whole_plant_balance_error =  
    (whole_carbon_after - whole_carbon_before) -  
    (npp_daily + unmet_storage_deficit)
```

Expected value:

```
whole_plant_balance_error    0
```

The `unmet_storage_deficit` term is included because when negative NPP exceeds available storage, the allocation routine reports the unpaid carbon demand rather than allowing storage to become negative.

7. One-year reference test

A one-year reference simulation was run with:

```
npp_rate = 3.5  
initial leaf = 1.0  
initial root = 0.8  
initial sapwood = 10.0  
initial heartwood = 20.0  
initial storage = 0.0
```

The main result after 365 days was:

Variable	Initial	Final
Leaf mass	1.000	2.121
Root mass	0.800	2.045
Sapwood mass	10.000	10.670
Height	13.136	13.190
Storage	0.000	0.466
Leaf-root residual	0.200	0.076

Pipe-model residual -12.360 -0.425

This behavior is consistent with the intended gradual allocation logic:

- leaf, root, and sapwood increased;
- root increased proportionally more than leaf, reducing the initial leaf-root imbalance;
- the pipe-model residual became much less negative, indicating movement toward pipe-model consistency;
- storage began to accumulate once daily structural demand became lower than daily NPP input;
- carbon-accounting errors remained near numerical precision, approximately 10^{-15} .

The annual carbon balance also closed: annual NPP was approximately equal to structural growth plus the increase in storage.

8. Unit tests performed

A separate Fortran unit-test driver was created:

`test_storage_allocation_unit_cases.f90`

The goal was to test expected behavior in simple, interpretable cases. All tests passed.

Test	Scenario	Expected behavior	Result
A	Zero NPP and zero storage	No structural growth should occur; storage remains zero.	PASS
B	Positive NPP with imbalanced plant	Structural pools grow and allometric residuals improve.	PASS
C1	Equilibrated plant with background growth enabled	The plant can continue balanced structural growth.	PASS

Test	Scenario	Expected behavior	Result
C2	Equilibrated plant with background growth disabled	Structural growth is near zero and carbon accumulates in storage.	PASS
D	Positive initial storage and zero NPP	Storage can support structural growth when demand exists.	PASS
E1	Negative NPP with enough storage	Storage is consumed without creating an unmet deficit.	PASS
E2	Negative NPP exceeding storage	Storage goes to zero and the remaining deficit is reported.	PASS
F	Very high storage and demand	Daily allocation is constrained by <code>max_allocation_fraction</code> .	PASS

Each unit test also checked:

- no negative structural pools;
- no negative storage;
- no negative structural increments;
- structural balance error near zero;
- storage balance error near zero;
- whole-plant balance error near zero.

9. Sensitivity-test design

A broader sensitivity-test driver was then created:

`test_storage_allocation_sensitivity.f90`

The test matrix included:

Dimension	Values tested
NPP rate	-0.5, 0.0, 0.5, 3.5, 8.0
Initial storage	0.0, 0.5, 5.0

Dimension	Values tested
Allometric adjustment time	30, 365, 730 days
Maximum daily allocation fraction	0.001, 0.005, 0.02
Background-growth mode	enabled, disabled
Trait combinations	base, low SLA, high SLA, dense wood
Initial plant states	original imbalanced, equilibrated, leaf-rich, root-rich, sapwood-rich

This resulted in:

$5 \times 3 \times 3 \times 3 \times 2 \times 4 \times 5 = 5400$ scenarios

Each scenario was run for:

5 years = 1825 daily timesteps

The output was saved in:

storage_allocation_sensitivity_summary.csv

10. Sensitivity-test results

All 5400 sensitivity scenarios completed successfully.

Diagnostic	Result
Total scenarios	5400
Passed scenarios	5400
Failed scenarios	0
Maximum absolute structural balance error	1.11×10^{-16}
Maximum absolute storage balance error	0.00
Maximum absolute whole-plant balance error	2.59×10^{-14}
Minimum living pool observed	0.8

These results indicate that the gradual storage allocation routine is numerically stable across the tested parameter space. No NaN, Inf, negative pool, negative storage, or carbon-accounting failure was detected.

10.1 Storage accumulation

Although all scenarios passed numerically, the sensitivity tests revealed an important biological/modeling behavior: storage can accumulate substantially when structural demand is low relative to NPP input.

Across the 5400 scenarios:

Storage criterion	Number of scenarios
Maximum storage fraction > 0.30	680
Maximum storage fraction > 0.50	216

This pattern was strongly associated with the absence of background balanced-growth demand:

Background-growth mode	Scenarios with storage fraction > 0.30	Scenarios with storage fraction > 0.50
Enabled	45	0
Disabled	635	216

This confirms that if allocation depends only on allometric correction deficits, an already equilibrated plant can accumulate carbon in storage without converting it into structural biomass. This is numerically valid but may be biologically undesirable unless storage accumulation is explicitly intended and constrained by additional processes such as reserve turnover, storage respiration, or sink limitation.

10.2 Maximum-storage case

The maximum storage fraction observed was approximately:

`max_storage_fraction = 0.649`

This occurred under the following conditions:

Parameter	Value
Background growth	Disabled
NPP rate	8.0
Initial storage	5.0
Allometric adjustment time	365 days
Maximum daily allocation fraction	0.001
Initial state	Approximately equilibrated
Trait case	Low SLA strategy
Cumulative NPP over 5 years	40.0
Cumulative structural allocation	0.0
Final storage	45.0

This is an expected result for an equilibrated plant with background growth disabled: because there is no allometric deficit, there is no structural demand, so incoming NPP remains in storage.

10.3 Interpretation of structural allocation fractions

Some scenarios produced structural allocation fractions greater than 1 when expressed relative to positive NPP alone. This is not necessarily an error because positive initial storage can also finance structural growth. Therefore, `structural_fraction_of_positive_npp` should be interpreted as a diagnostic index, not as a strict carbon-use efficiency.

11. Interpretation of the tests

The tests support three main conclusions.

First, the gradual storage allocation routine is numerically robust under the tested conditions. It conserves carbon to numerical precision, avoids negative pools, respects the daily allocation limiter, and handles negative NPP by explicitly reporting unmet storage deficits.

Second, the background balanced-growth term is functionally important. Without it, plants that are close to allometric equilibrium may stop structural growth even under positive NPP, causing storage to accumulate. This behavior is consistent with the equations, but it may not be biologically desirable for normal growth conditions. Therefore, this term should be interpreted as a balanced structural growth demand rather than an arbitrary basal growth term.

Third, the tests validate the numerical basis of the allocation scheme, but they do not by themselves prove that the chosen parameter values are biologically optimal. Parameters such as `allometric_adjustment_days`, `max_allocation_fraction`, and the background growth timescales should still be treated as sensitive model parameters and evaluated against expected plant growth behavior or empirical benchmarks when possible.

12. Implications for adding growth respiration

The current scheme treats `carbon_to_allocate` as carbon converted into structural growth. Growth respiration has not yet been included. Because the allocation routine is now carbon-conservative without growth respiration, the next step can be implemented and tested cleanly.

The recommended formulation is to define structural growth as the net biomass increment and compute the total carbon cost of that growth using a growth efficiency. For example, if growth respiration is 25% of the total carbon cost, then growth efficiency is 0.75:

`growth_efficiency = 0.75`

For a requested structural increment:

`structural_growth = delta_leaf + delta_root + delta_sapwood`

The total carbon withdrawn from storage should be:

`total_growth_cost = structural_growth / growth_efficiency`

Growth respiration would then be:

`growth_respiration = total_growth_cost - structural_growth`

After this modification, the storage balance should become:

`carbon_storage_after = carbon_storage_before + npp_daily - total_growth_cost`

rather than:

`carbon_storage_after = carbon_storage_before + npp_daily - structural_growth`

This distinction is essential. With growth respiration included, the carbon withdrawn from storage will be larger than the carbon incorporated into biomass.

13. Recommended next validation step

After implementing growth respiration, the same unit tests and the same 5400-scenario sensitivity matrix should be rerun. The expected changes are:

1. Structural growth should be lower for the same amount of available storage.
2. Storage should decline faster because it pays both structural growth and growth respiration.
3. A new diagnostic should verify that:

`total_carbon_withdrawn_from_storage = structural_growth + growth_respiration`

4. Whole-plant carbon balance should include growth respiration as an explicit carbon loss:

`change_in_structural_carbon + change_in_storage + growth_respiration = npp_daily`

or, depending on the model-level definition of NPP:

`change_in_structural_carbon + change_in_storage = NPP_after_growth_respiration`

The key conceptual decision before coding is whether the input to this allocation routine represents carbon before or after growth respiration. The recommended approach is to pass carbon available before growth respiration, let the allocation routine compute structural growth and growth respiration, and report both explicitly.

Final storage fraction by NPP, with background balanced-growth demand enabled.

Figure 1: Final storage fraction by NPP, with background balanced-growth demand enabled.

Final storage fraction by NPP, with background balanced-growth demand disabled.

Figure 2: Final storage fraction by NPP, with background balanced-growth demand disabled.

14. Summary

The gradual storage allocation scheme provides a more biologically plausible daily allocation framework than rigid instantaneous allocation because it allows carbon storage, sink limitation, gradual allometric correction, and bounded daily growth. The validation tests show that the current implementation conserves carbon and behaves robustly across a broad range of parameter combinations.

The most important modeling insight from the tests is that background balanced-growth demand is necessary to avoid excessive storage accumulation when plants are already close to allometric equilibrium. Therefore, this term should be retained or replaced by a more mechanistic sink-demand formulation before moving to full growth respiration.

The code is now in a suitable state to proceed to the next development step: adding growth respiration with explicit carbon-accounting diagnostics.

Appendix A. Diagnostic figures

The following plots were generated from the sensitivity-test output and can be used for visual inspection of the storage and allometric responses.

Fraction of positive NPP converted into structural biomass, with background balanced-growth demand enabled.

Figure 3: Fraction of positive NPP converted into structural biomass, with background balanced-growth demand enabled.

Fraction of positive NPP converted into structural biomass, with background
balanced-growth demand disabled.

Figure 4: Fraction of positive NPP converted into structural biomass, with
background balanced-growth demand disabled.

Final pipe-model residual by initial state case.

Figure 5: Final pipe-model residual by initial state case.